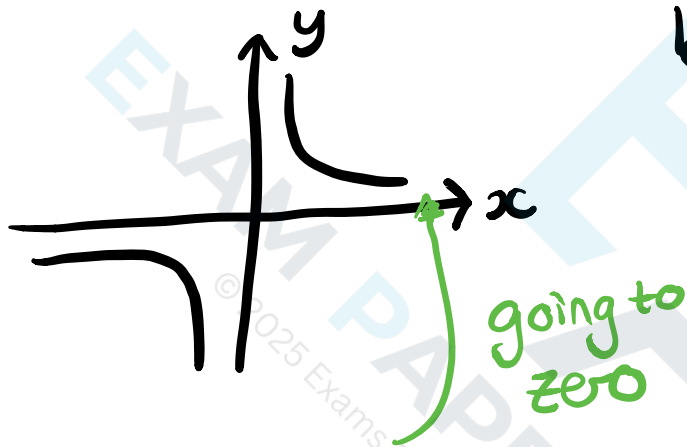


Limits of rational polynomials

Basic skills

Q1. $y = \frac{1}{x}$ & reciprocal graph

a)



b) $x=0$ is an undefined asymptote
 $y=0$ is a limiting asymptote

c) $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$

d) $\lim_{x \rightarrow 0} \frac{1}{x}$ doesn't exist

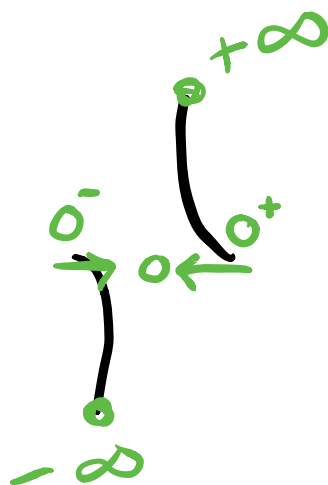
as $\frac{1}{x} \rightarrow +\infty, -\infty$

depending on if

$x \rightarrow 0^+, x \rightarrow 0^-$

$\uparrow$
0 from
+ve

$\uparrow$
0 from
-ve



Q2.

a) $y = \frac{4x^2 - x - 2}{x^2} = \frac{4x^2}{x^2} - \frac{x}{x^2} - \frac{2}{x^2}$

$$= 4 - \frac{1}{x} - \frac{2}{x^2}$$

b) $x \rightarrow \infty \quad y \rightarrow 4 - 0 - 0 = 4$

c) $x \rightarrow 0 \quad y \rightarrow -\infty$ as $\frac{1}{x} \rightarrow \infty, \frac{2}{x^2} \rightarrow \infty$

Competency skills

Q3.

a) $\frac{0}{0}$ Problem

b) $\frac{\infty}{\infty}$ Problem

c) $\frac{\infty}{0}$ Problem

Q4.

a) $\lim_{x \rightarrow 0} \frac{x^2 - x}{x} = \lim_{x \rightarrow 0} (x - 1) = 0 - 1 = -1$

smallest Power

b) $\lim_{x \rightarrow \infty} \frac{x^3 + x^2}{x^3} = \lim_{x \rightarrow \infty} (1 + \frac{1}{x}) = 1 + 0 = 1$

largest Power

$$c) \lim_{x \rightarrow \infty} \frac{x^3 - 2x^4 + x + 2}{x^4} = \lim_{x \rightarrow \infty} \left(\frac{1}{x} - 2 + \frac{1}{x^3} + \frac{2}{x^4} \right)$$

$$= 0 - 2 + 0 + 0 = -2$$

$$d) \lim_{x \rightarrow 0} \frac{x+2}{x^2-1} = \frac{2}{-1} = 2$$

no problem

$$e) \lim_{x \rightarrow \infty} \frac{(x+1)^2}{2x(x+1)} = \lim_{x \rightarrow \infty} \frac{x^2 + 2x + 1}{2x^2 + 2x}$$

$$= \lim_{x \rightarrow \infty} \frac{1 + \frac{2}{x} + \frac{1}{x^2}}{2 + \frac{2}{x}} = \frac{1+0+0}{2+0} = \frac{1}{2}$$

$$f) \lim_{x \rightarrow \infty} \frac{(3x-1)^2}{2x^3} = \lim_{x \rightarrow \infty} \frac{9x^2 - 6x + 1}{2x^3}$$

$$= \lim_{x \rightarrow \infty} \left(\frac{9}{2x} - \frac{3}{x^2} + \frac{1}{2x^3} \right) = 0 - 0 + 0 = 0$$

$$g) \lim_{y \rightarrow \infty} \frac{5y^3 - 7y + 5}{2y^2 + 5} = \lim_{y \rightarrow \infty} \frac{5y - \frac{7}{y} + \frac{5}{y^2}}{2 + \frac{5}{y^2}}$$

$$= \infty$$

Q5. $\lim_{x \rightarrow 0} \frac{(2x - \frac{2}{x})^2}{x^2} = \lim_{x \rightarrow 0} \frac{4x^2 - 8 + \frac{4}{x^2}}{x^2}$

$$= \lim_{x \rightarrow 0} \left(4 - \frac{8}{x^2} + \frac{4}{x^4} \right) = \infty$$

Advanced skills

Q6. $\lim_{x \rightarrow 9} \frac{(x-9)^2}{81-x^2} = \lim_{y \rightarrow 0} \frac{y^2}{81-(y+9)^2}$

$y = x - 9$

$$= \lim_{y \rightarrow 0} \frac{y^2}{-y^2 - 18y} = \lim_{y \rightarrow 0} \frac{y}{-y - 18} = \frac{0}{-18}$$

$$= 0$$

Q7. $\lim_{x \rightarrow 0} \frac{\sin x}{\cos x} = \frac{0}{1} = 0$

AS no issue / Problem as 0 on top and no 0 or ∞ on bottom

$\frac{0}{n}$ is not a Problem

Q8.

a) $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{6x^2 + 5}{2x - 5} = \lim_{x \rightarrow \infty} \frac{6x + \frac{5}{x}}{2 - \frac{5}{x}}$
 $= \infty$

Handwritten notes: Red arrows point from the $\frac{5}{x}$ terms to ∞ and 0 respectively.

b) $\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} \frac{2x - 5}{6x^2 + 5} = \lim_{x \rightarrow -\infty} \frac{\frac{2}{x} - \frac{5}{x^2}}{6 + \frac{5}{x^2}}$
 $= 0$

Handwritten note: Red arrow points from the $\frac{5}{x^2}$ term to 0 .

c) $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{6x^2 + 5}{2x - 5} = \frac{5}{-5} = -1$

$\lim_{x \rightarrow 0} \frac{2x - 5}{6x^2 + 5} = \frac{-5}{5} = -1$

d) Yes it doesn't exist

